

# Jack Xiuqi Zhang

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## EDUCATION

### Imperial College London

MEng Computing

London, UK

2023 – 2027

- Achieved High First Class in first, second, and third year.
- Third-year modules: Introduction to ML, Maths for ML, Deep Learning, Computer Vision, Robotics (led team to 2nd in cohort — object-avoiding path planning), Optimisation.

### Alice Smith School

A Levels

Kuala Lumpur, Malaysia

2019 – 2023

## EXPERIENCE

### US-based Hedge Fund

Software Engineer (Industrial Placement)

Mar. 2026 – Present

- Built a production LLM chat agent over a **130k-document news corpus** (langchain deepagents, Manticore search, OpenUI) — prototyped in 3 days, now a **20-user pilot**; a PM called it a “**game changer**”.
- Designed and implemented the **evaluation harness** (LangSmith; recall and recency evals from real user feedback); trimmed search-result metadata, cutting **tokens per search by 90% and total tokens per query by 70% with 0% recall impact**.
- Deployed an **MCP server** (index / search / get\_document) so users can connect their own Claude/GPT; debugged production issues on **Kubernetes** (Helm, kubectl, Datadog) and fixed an agent-stopping bug by gating deepagents auto-compaction output.
- Benchmarked a FastAPI + Postgres service (Locust): fixed connection-pool, worker, and row-lock bottlenecks — **1000 RPS write at p95 577ms**; deployed indexes improving REST reads by **129.9%** and GraphQL by **33.6%**.
- Added **Microsoft Entra ID bearer-token auth** to a service exposed via the cluster edge — previously callable by any machine that could reach it (closed a cluster-wide exposure).

### Goodnotes

London, UK

Machine Learning and Software Engineer Intern

Jun. 2025 – Sep. 2025

- Owned end-to-end AI-powered outline generation feature from PRD to production, targeting **2M+ users**.
- Implemented a Python processing pipeline within the existing **FastAPI + AWS Bedrock** backend; self-taught **Swift/iOS** and built the production-ready frontend.
- Fixed a critical JSON bug affecting **50k+ daily users**, reducing error rates from **40% to near-zero**.
- Enhanced CI/CD reliability by adding **pytest** retries, lowering pipeline failure rate from **30% to <5%**.

### 15 AI

Founding Engineer

Mar. 2025 – Apr. 2026

- Built a search engine over scraped Key Opinion Leader data (contacts queryable by product/tag) and an Astro-based content site; owned product development end-to-end through rapid prototyping as the company iterated on direction.

### KEATH.AI

Software Engineer and Prompt Engineer Intern

Jun. 2024 – Sep. 2024

- Engineered a comparative marking system with education specialists, improving AI grading accuracy by **50%**.
- Built a **RAG-based chatbot** (Flask + OpenAI API), reducing staff workload by automating Q&A.

## PROJECTS

**PROBE — Research Benchmark** | Python, coding agents | [github.com/LemonPhase/PROBE](https://github.com/LemonPhase/PROBE)

2026 – Present

- Open-source research benchmark on how repository documentation structure affects coding-agent performance; SWE-bench-derived bugfix instances with runtime validation.

**WACC Compiler** | Scala, Intel x86

Jan. 2025 – Apr. 2025

- Led 4-person team to develop a **full-featured compiler** (lexing, parsing, semantic checks, Intel x86 code generation) with external **C FFI** integration.
- Built a **100+ test suite** and GitLab CI for automated testing and static analysis.

**Pintos Operating System** | C

Sep. 2024 – Dec. 2024

- Directed team of 4 implementing **threading, system calls, and virtual memory** (clock eviction, lazy loading, page sharing); designed a **priority donation algorithm** preventing priority inversion in the kernel.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, Java, Scala, Swift, TypeScript, SQL

**AI/LLM Engineering:** LangChain (deepagents), LangSmith evals, MCP, RAG/vector search (Manticore), prompt engineering

**Frameworks & Libraries:** FastAPI, PyTorch, TensorFlow, React, SwiftUI/UIKit, NumPy, Pandas

**Developer Tools:** Git, Linux/Bash, Docker, AWS, Kubernetes (Helm, kubectl), Locust, GDB